Northeastern University
College of Engineering
Department of Electrical & Computer Engineering

EECE2310 – Introduction to Digital Design and Computer Architecture

Homework 1

Submission Instructions

- Homework assignments must be done by each student individually (no groups are allowed).
- For the programming problems (RISC-V assembly):
    1. Your code must be well commented by explaining what the lines of your program do. Have at least one comment for every 4 lines of code.
    2. At the beginning of your source code files (only if a separate source code file is required by the question) write your full name, and NU student ID.
    3. If required by the question, test your code using the RARS assembler.
- For non-programming problems, include explanation of all steps of your answers/calculations not only the final results.
- Submit the following to the homework assignment page on Canvas:
    1. Your homework report developed by a word processor (e.g., Microsoft Word) and submitted as one PDF file. Any handwritten answers will not be accepted. For answers that require drawing of diagrams, one option is to use Lucid (check the *Creating Diagrams with Lucid* document on the Miscellaneous Resources page on Canvas). The report includes the following (depending on the assignment contents):
        a. Answers to the non-programming problems that show all the details of the steps you follow to reach these answers.
        b. A summary of your approach to solving the programming problems.
        c. The screen shots of the sample runs of your programs.
    2. If required by any of the questions, your well-commented programs source code files (i.e., the .asm files).

Do NOT submit any files (e.g., the PDF report file and the programs source code files) as a compressed (zipped) package. Rather, upload each file individually.
Do NOT post solutions to any assignment or projects on public repositories like GitHub. Sharing your work in this way can compromise the learning experience for other students and could be considered academic dishonesty.

Note: You can submit multiple attempts for this homework, however, only what you submit in the last attempt will be graded (i.e., all required reports and files must be included in this last attempt).

Q1 (10 Points)

The following are functions of x, y, and z. Find the **complement** of the functions by applying DeMorgan's theorems as many times until it cannot be applied anymore:

(a) $F1 = x'yz' + x'y'z$ (b) $F2 = x(y'z' + yz)$

Q2 (10 Points)

Express the **complement** of the following functions in sum-of-minterms form:

(a) $F(A,B,C,D) = \Sigma m(2, 4, 7, 10, 12, 14)$

(b) $F(x, y, z) = \Pi M(3, 5, 7)$

Q3 (10 Points)

Express the following unsigned numbers in decimal (show all the steps):

(a) $(10110.0101)_2$ (b) $(16.5)_{16}$

(c) $(26.24)_8$ (d) $(DADA.B)_{16}$

Q4 (10 Points)

Convert the following unsigned **binary** numbers to **hexadecimal** and to **decimal** (show all the steps):

(a) 1.10010 (b) 110.010

By comparing their binary representation, explain why the decimal answer in (b) is 4 times that in (a).

Q5 (10 Points)

Convert decimal numbers +49, +29, -49, -29 to binary, using the signed-2's-complement representation with 8-bit storage. Then perform the binary equivalent of the following operations:

(a) (+29) + (-49) (b) (-29) + (+49) (c) (-29) + (-49).

Convert the binary results back to decimal and verify that they are correct.

Q6. (10 Points)

In an 8-bit 2's complement binary system, what is the result of each of the following subtract operations? Is there any overflow? You must show how you detect overflow in the result only by verifying the **carry into the most significant bit and the carry out** of it.

(a)

```
  00101011
- 10110110
```

(b)

```
  01000001
- 10000010
```

Q7 (10 Points)

Express the following function as a sum of minterms (e.g., $F = \Sigma m(3, 6, 8)$ and as a product of maxterms (e.g., $F = \Pi M(3, 6, 8)$: (show all steps)

$F(A, B, C, D) = B'D + A'D + BD$

Q8 (10 Points)

Simplify the following Boolean functions, using *Karnaugh* maps:

(a) $F(A,B,C,D) = \Sigma(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$

(b) $F(A,B,C,D) = AB'C + B'C'D' + BCD + ACD' + A'B'C + A'BC'D$

Q9 (10 Points)

Simplify the following Boolean function using *Karnaugh* map and draw the digital circuit to implement the function using **only NAND** gates:

$$F(A,B,C,D) = \Sigma(0, 1, 2, 3, 6, 10, 11, 14)$$

Q10 (10 Points)

Given four binary inputs *ABCD*, design a digital function with an output *Z* where *Z* is only 1 if the decimal value of the 4-bit input is dividable by 3 or by 4 without remainder (*Note: Z* is 1 for input value zero).

a. Provide the truth table for this function.
b. Use Karnaugh map to obtain the <u>minimal</u> sum-of-products Boolean expression of *Z*.

No need to draw the digital circuit of this function.